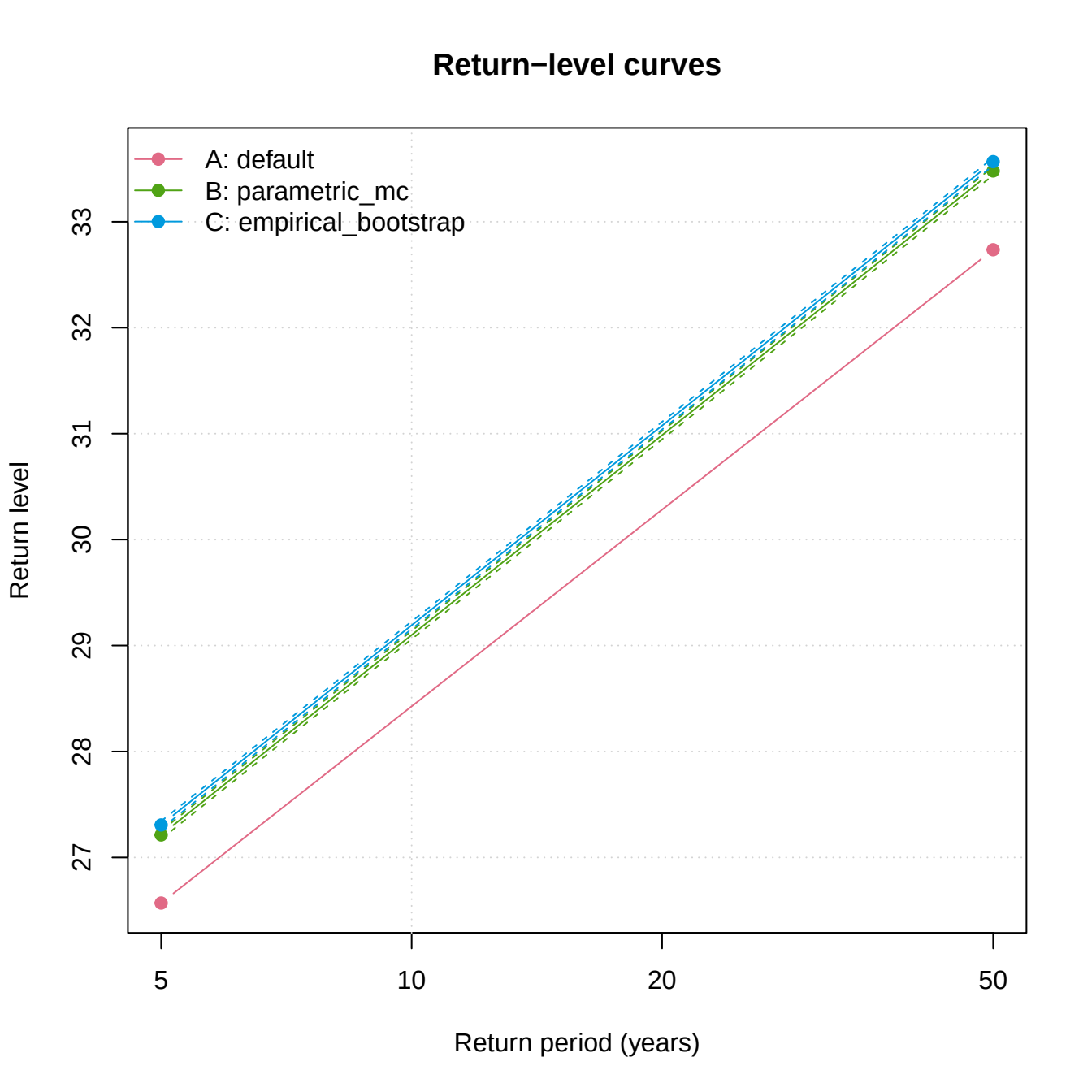
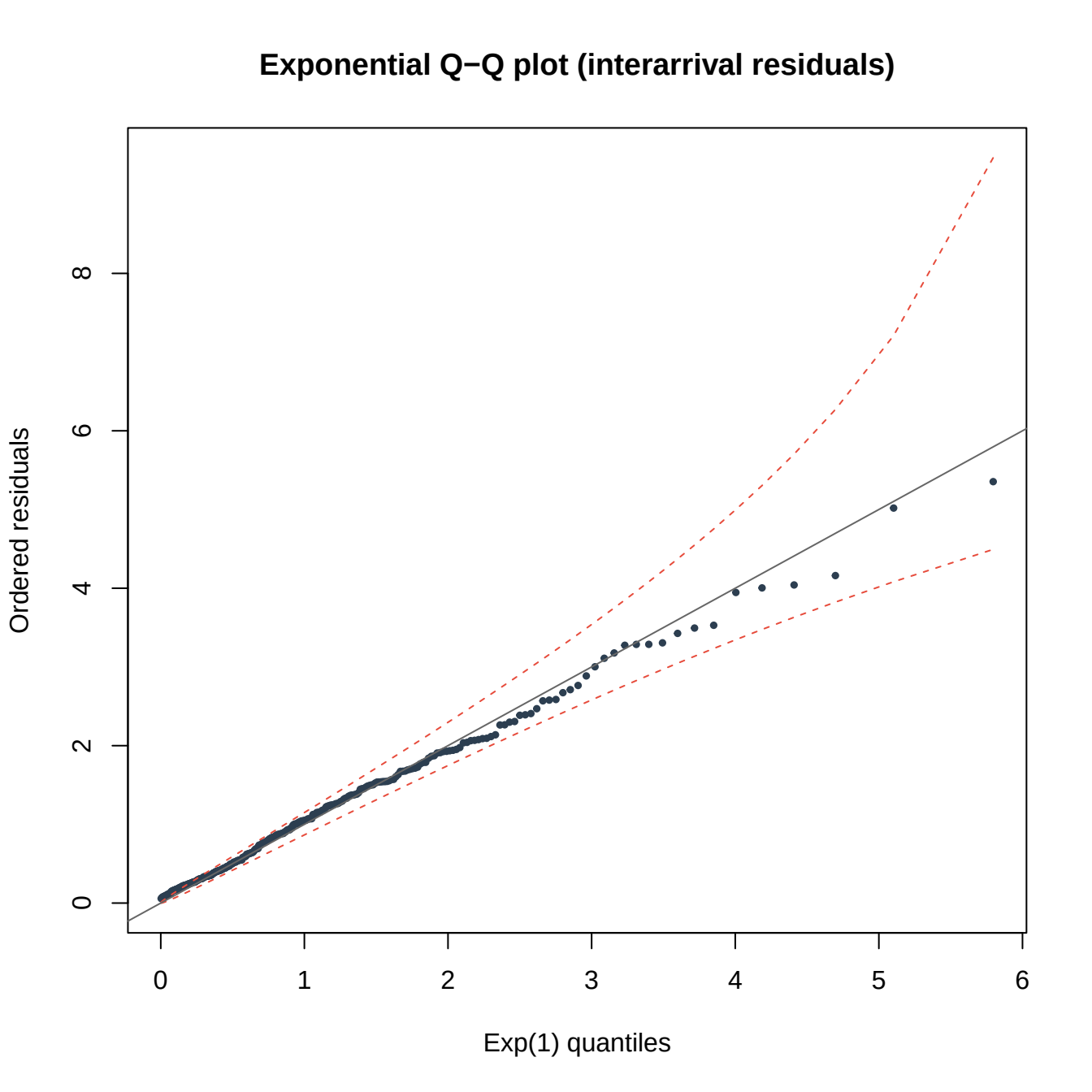
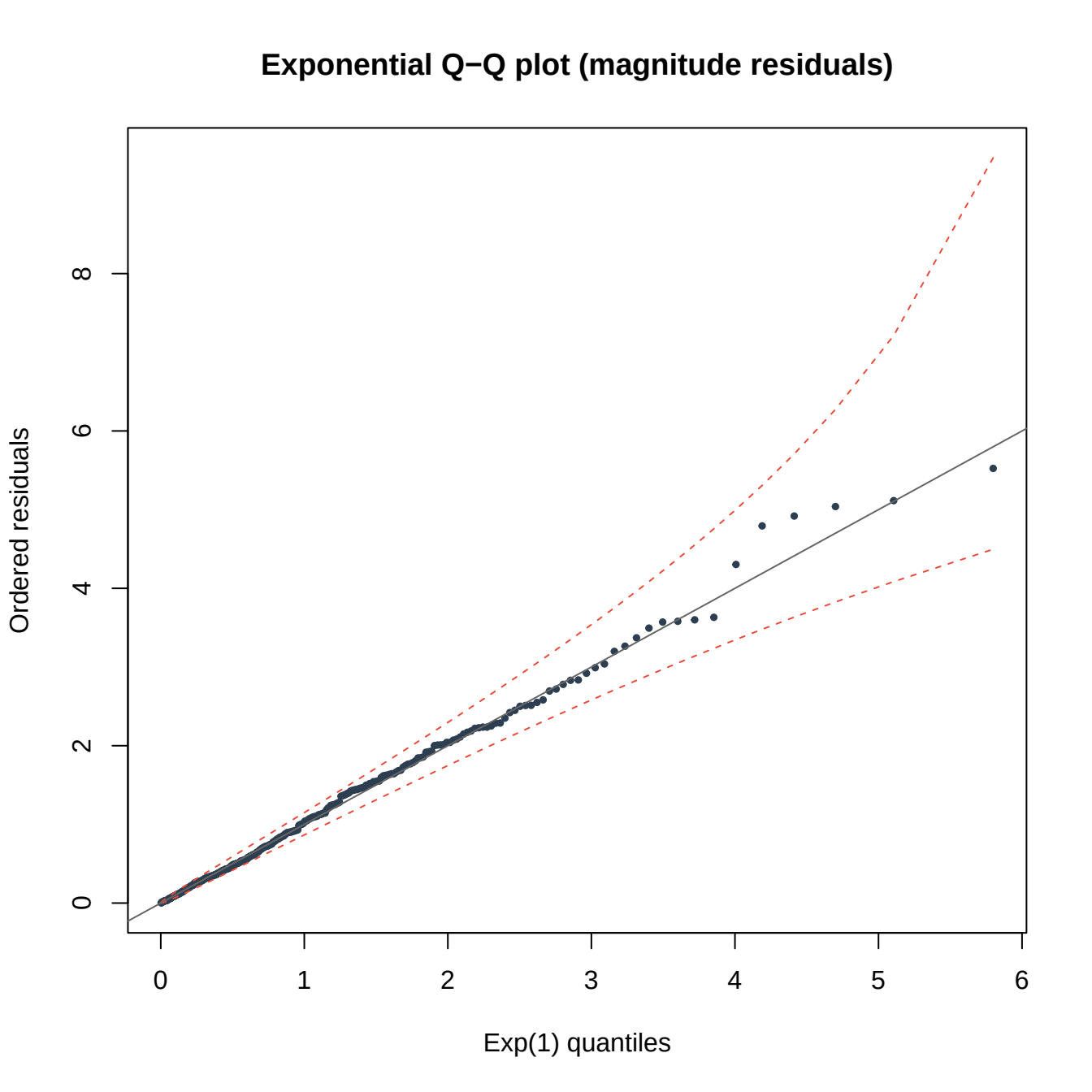
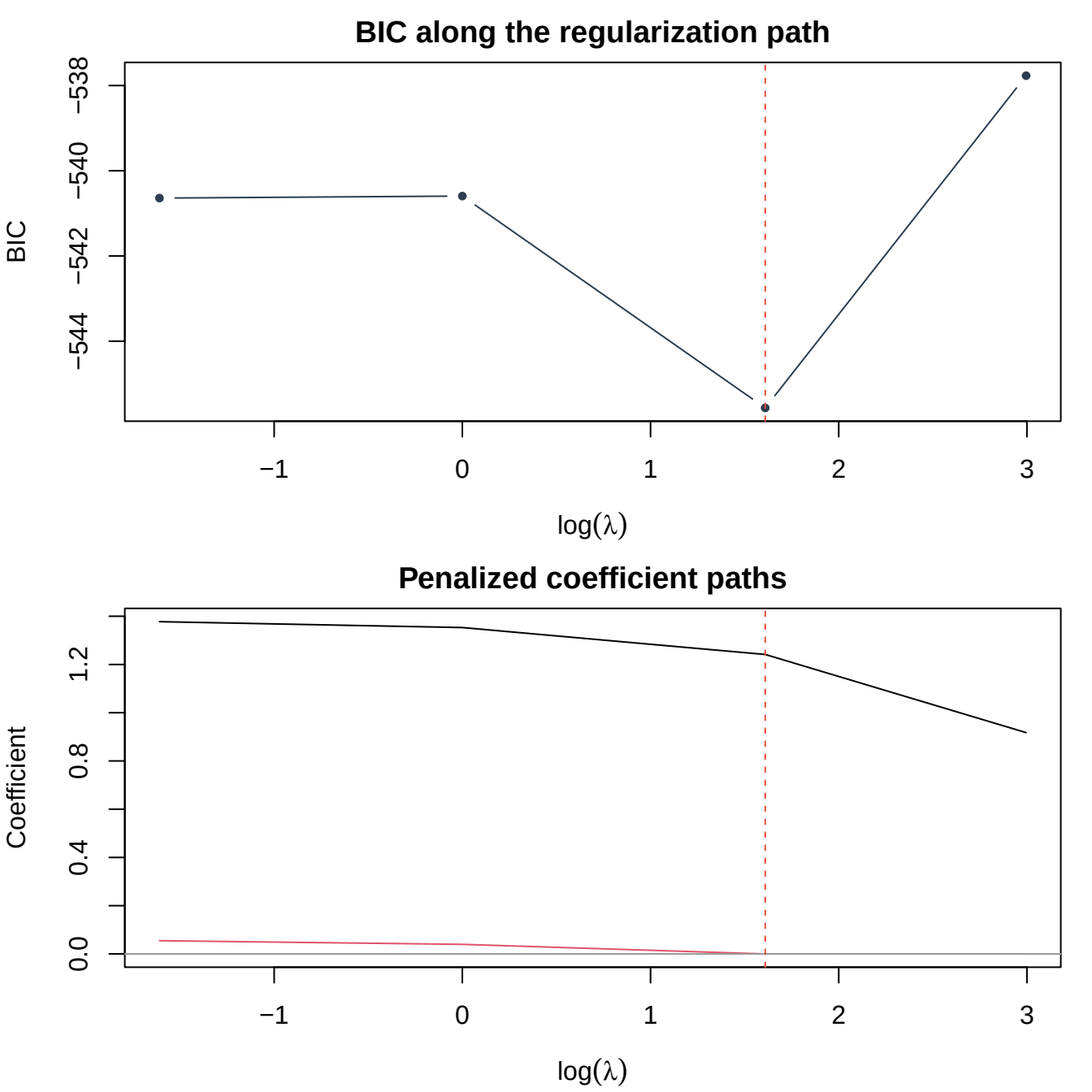
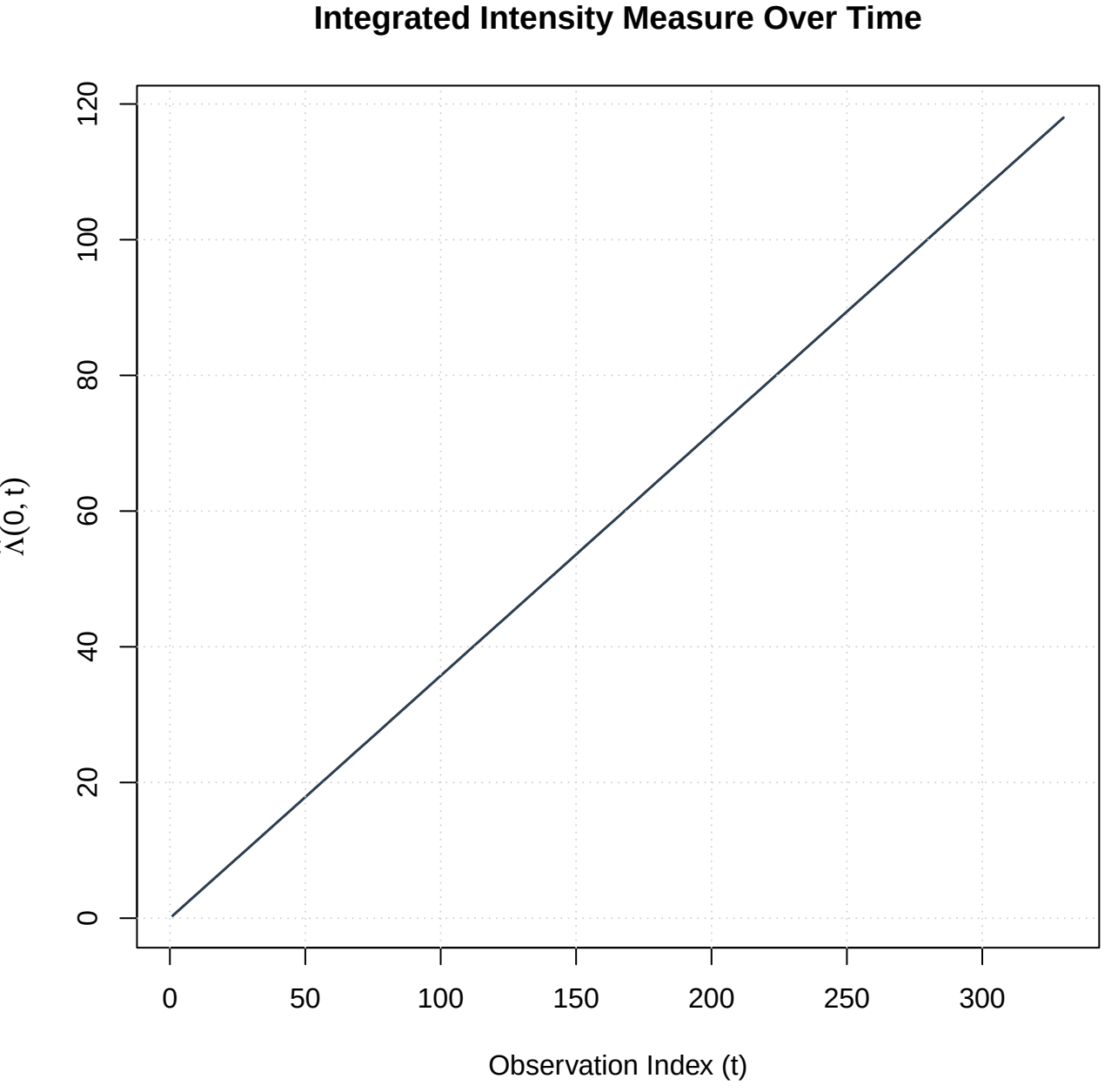
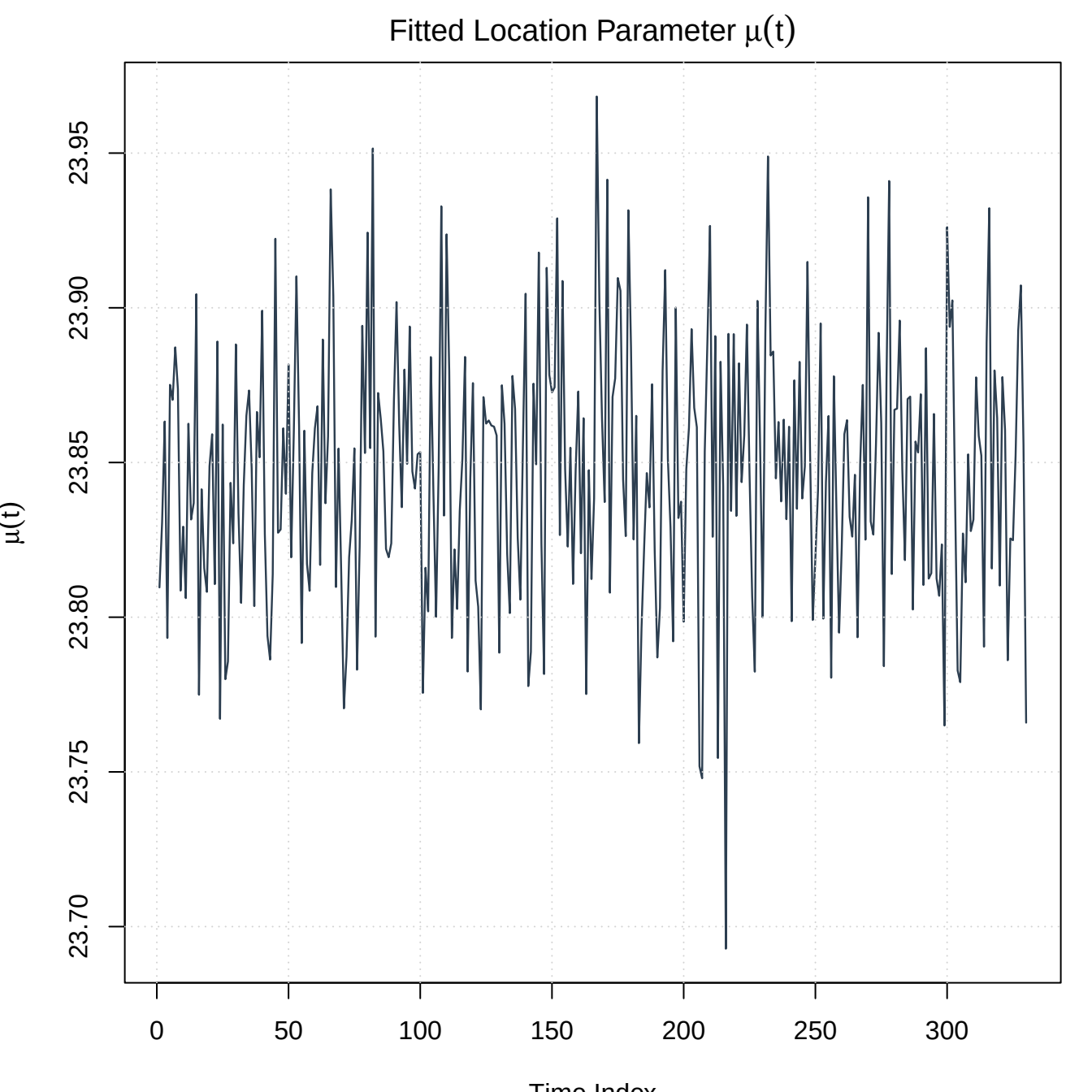

Return−level curves
A: default
B: parametric_mc
C: empirical_bootstrap
Return level
27
28
29
30
31
32
33
5
10
20
50
Return period (years)

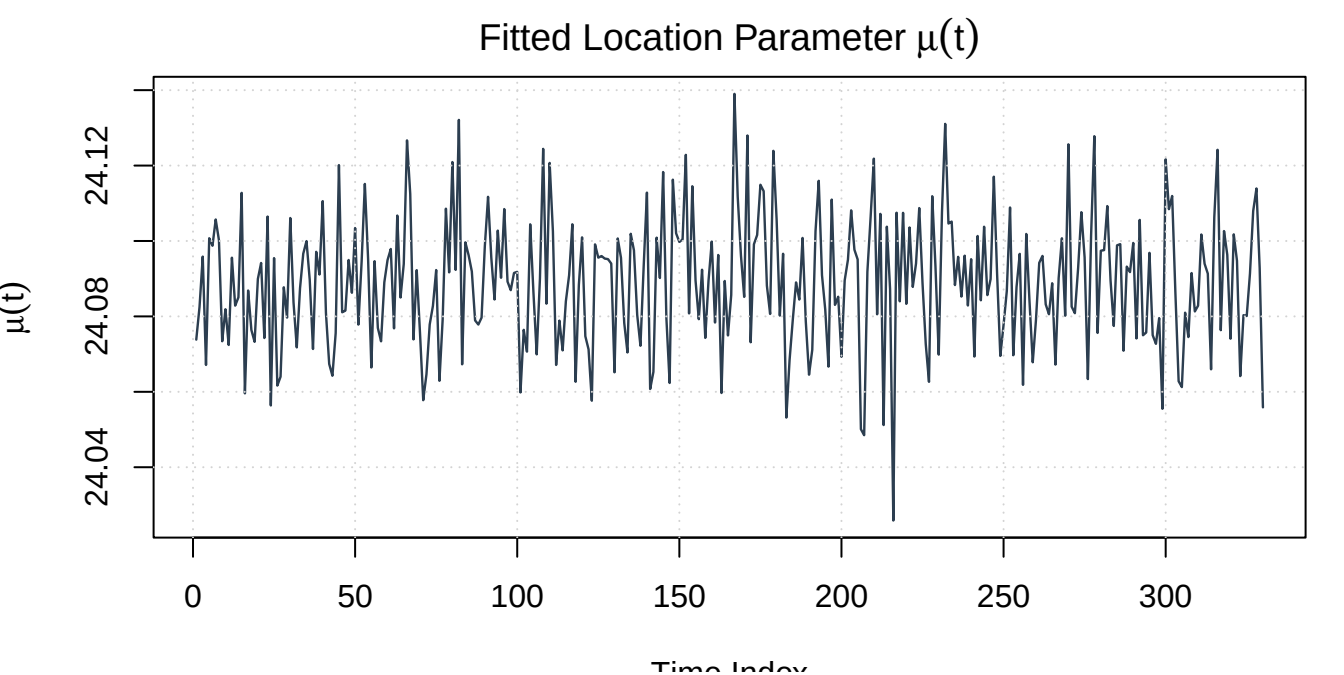

Exponential Q−Q plot (interarrival residuals)
Ordered residuals
0
2
4
6
8
0
1
2
3
4
5
6
Exp(1) quantiles

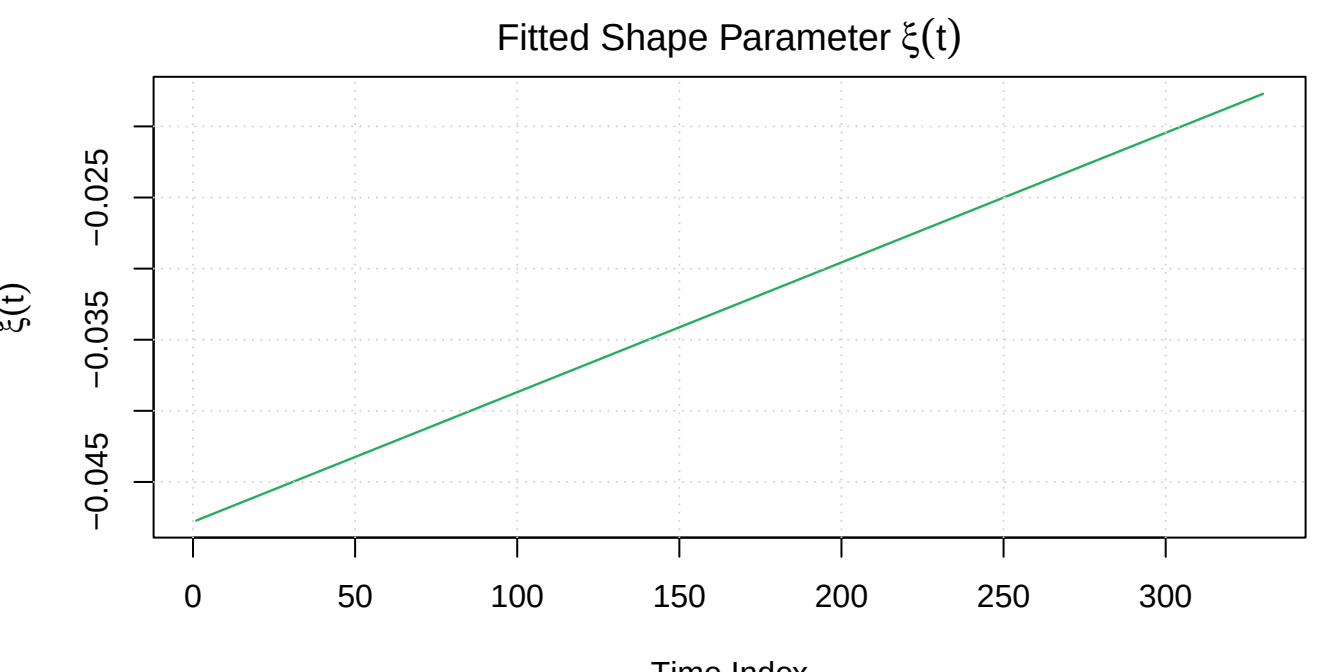

Exponential Q−Q plot (magnitude residuals)
Ordered residuals
0
2
4
6
8
Exp(1) quantiles
0
1
2
3
4
5
6

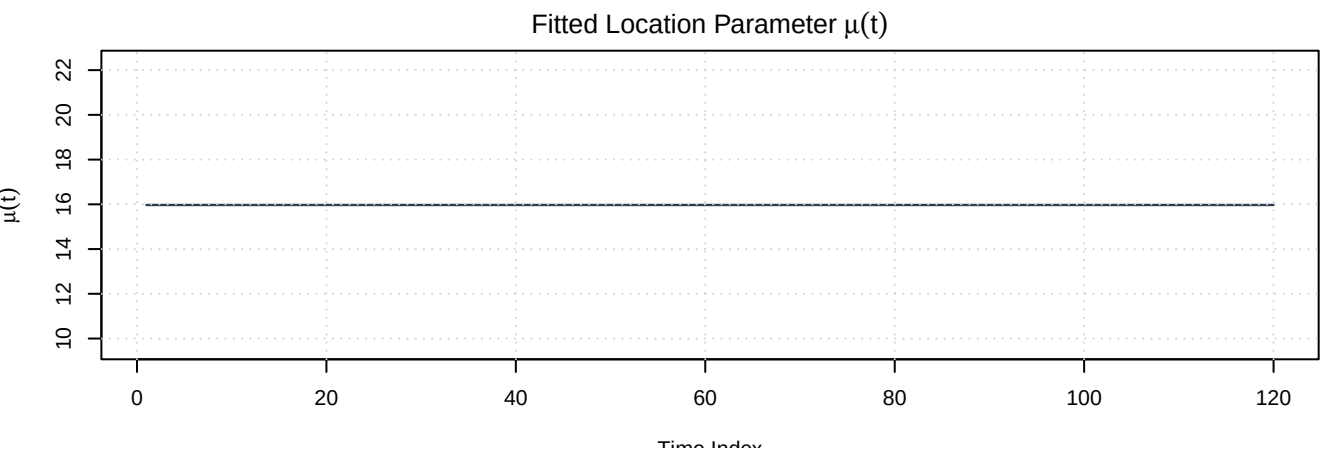

BIC along the regularization path
BIC
−538
−540
−542
−544
−1
0
1
2
3
log(λ)
Penalized coefficient paths
Coefficient
0.0
0.4
0.8
1.2
−1
0
1
2
3
log(λ)

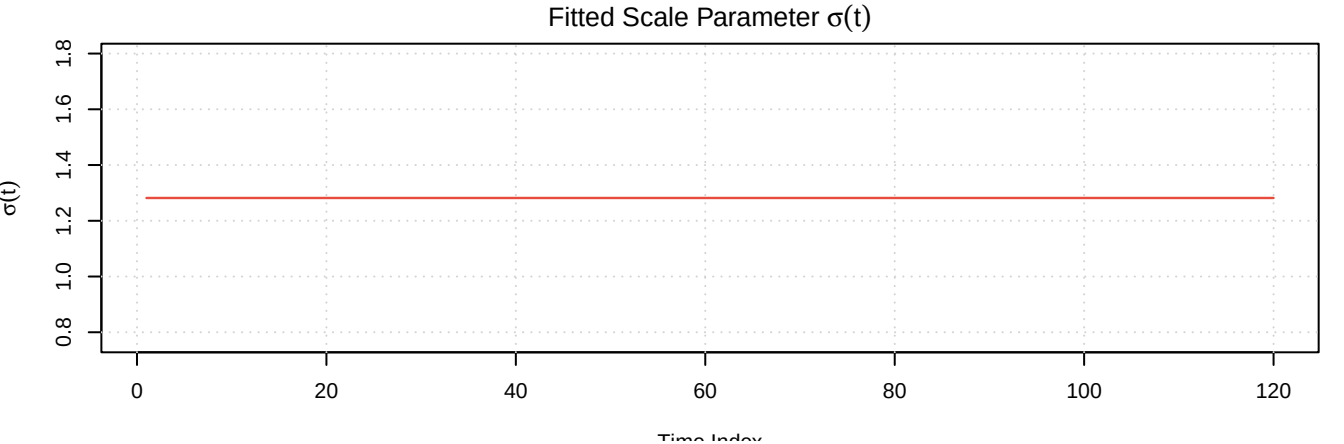

Integrated Intensity Measure Over Time
$\hat{\Lambda}(0, t)$
0
20
40
60
80
100
120
0
50
100
150
200
250
300
Observation Index (t)

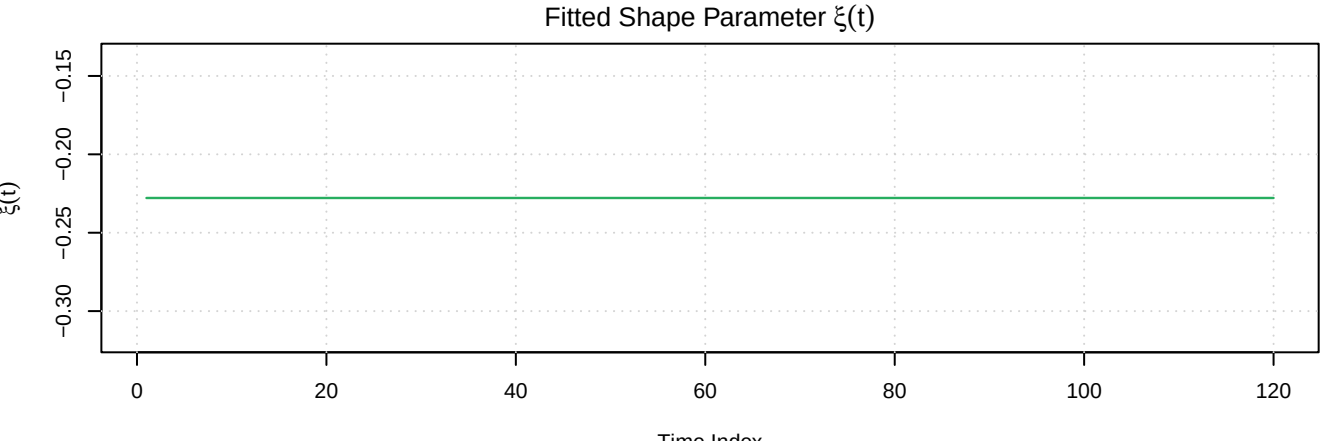

Fitted Location Parameter μ(t)
μ(t)
23.70
23.75
23.80
23.85
23.90
23.95
0
50
100
150
200
250
300
Time Index

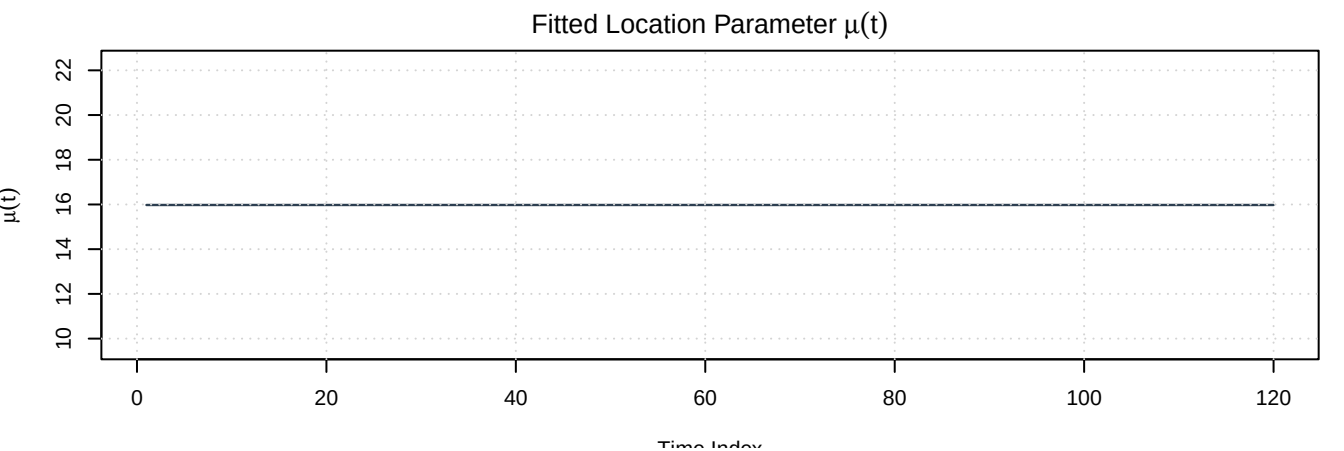

Fitted Location Parameter μ(t)
μ(t)
24.04
24.08
24.12
0
50
100
150
200
250
300
Time Index

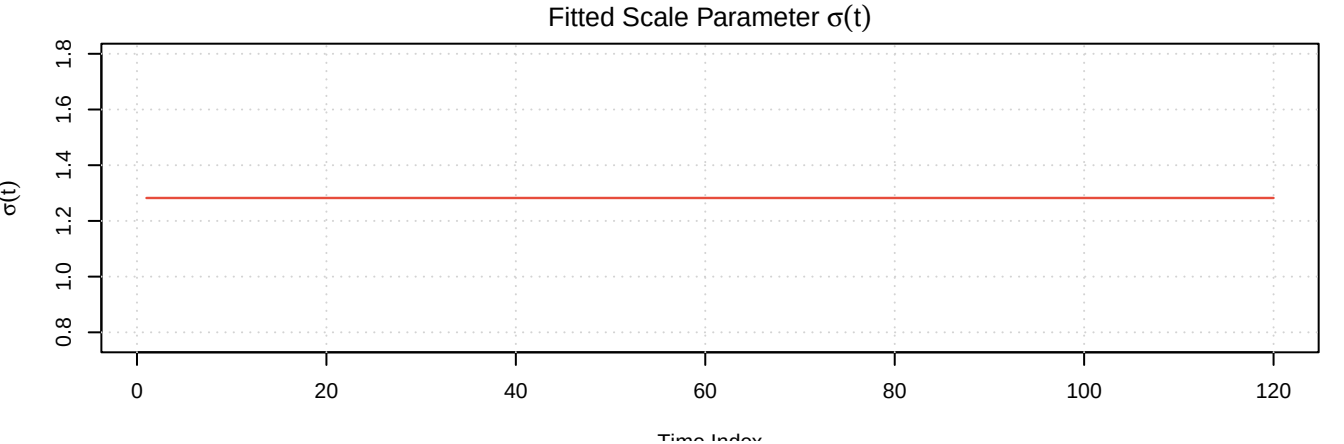

Fitted Shape Parameter ξ(t)
ξ(t)
−0.045
−0.035
−0.025
0
50
100
150
200
250
300
Time Index

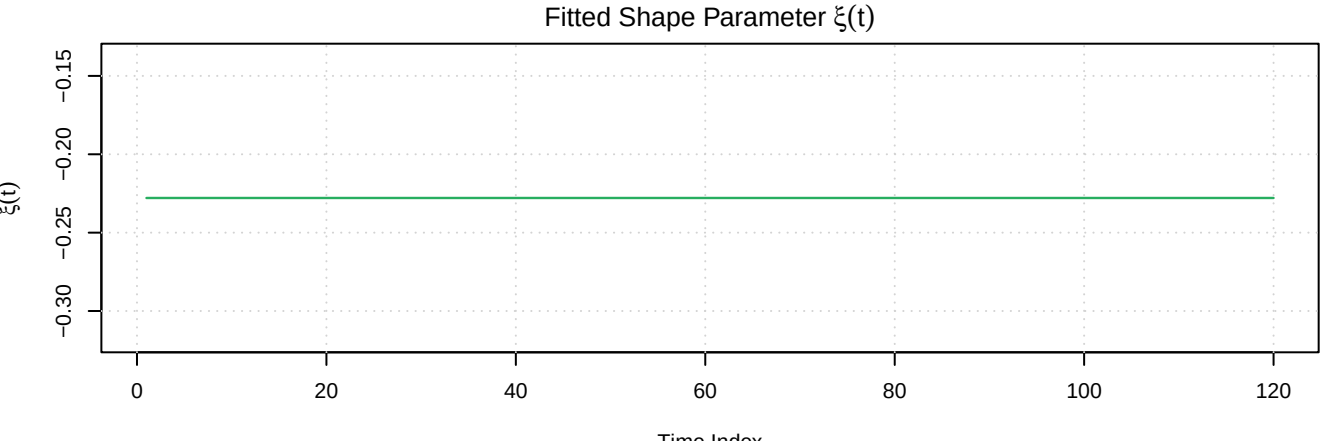

Fitted Location Parameter μ(t)
μ(t)
10
12
14
16
18
20
22
0
20
40
60
80
100
120


Fitted Scale Parameter σ(t)
σ(t)
0.8
1.0
1.2
1.4
1.6
1.8
0
20
40
60
80
100
120


Fitted Shape Parameter ξ(t)
ξ(t)
−0.30
−0.25
−0.20
−0.15
0
20
40
60
80
100
120


Fitted Location Parameter μ(t)
μ(t)
10
12
14
16
18
20
22
0
20
40
60
80
100
120


Fitted Scale Parameter σ(t)
σ(t)
0.8
1.0
1.2
1.4
1.6
1.8
0
20
40
60
80
100
120


Fitted Shape Parameter ξ(t)
ξ(t)
−0.30
−0.25
−0.20
−0.15
0
20
40
60
80
100
120